

No Evidence for Redshift Evolution of the $L_X - L_{UV}$ Relation from 2421 Quasars: A Critical Prerequisite for H_0 Measurements

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3 August 2026

ABSTRACT

We present a homogeneous analysis of the X-ray to ultraviolet luminosity relation ($\log L_X - \log L_{UV}$) for a sample of 2421 quasars from Lusso et al. 2020 (VizieR J/A+A/642/A150), spanning $0.1 < z < 5.0$. To avoid circularity we fit in flux-flux ($F_X - F_{UV}$) space in 4 narrow z -bins using D’Agostini (2005) likelihood with intrinsic scatter δ via MCMC (emcee), modeling $\log L_X = \gamma \log L_{UV} + \beta$. We find $\gamma = 0.6634 \pm 0.0055$, $\beta = 6.3344 \pm 0.16$, $\delta = 0.3167 \pm 0.0046$ dex. Testing evolution $\gamma(z) = \gamma_0 + mz$ gives $m = -0.002 \pm 0.007$, consistent with zero. A dedicated Monte Carlo simulation with 10,000 mocks demonstrates selection bias < 0.01 , an order of magnitude below our uncertainty. We find no evidence for evolution to $z = 5$, supporting the use of quasars as standardizable candles within current uncertainties, a critical prerequisite for Paper II where we will measure H_0 .

1 INTRODUCTION

The present expansion rate of the Universe, the Hubble constant H_0 , is in $> 5\sigma$ tension. Local distance-ladder measurements using Cepheids and SNe Ia give $H_0 = 73.04 \pm 1.04$ km s^{−1} Mpc^{−1} (Riess et al. 2022), while early-Universe inference from the CMB assuming Λ CDM gives $H_0 = 67.4 \pm 0.5$ km s^{−1} Mpc^{−1} (Planck Collaboration 2020). This Hubble tension (Di Valentino et al. 2021) motivates independent probes at $z > 2$ where SNe Ia are scarce.

Quasars can fill this gap via the ($\log L_X - \log L_{UV}$) relation, first reported by Tananbaum et al. (1979) and quantified by Avni & Tananbaum (1986). It links UV disc emission to X-ray coronal emission via Compton up-scattering, a fundamental prediction of accretion theory where $\gamma < 1$ reflects Compton cooling (Kubota & Done 2018).

Lusso et al. (2020) found $m = -0.05 \pm 0.02$ (2.5σ) with 2421 quasars but without $E(B - V) < 0.1$ and BAL cuts. Bisogni et al. (2021) found $m = -0.02 \pm 0.01$ with 130 quasars. In present work, After careful selection removing dust reddening ($E(B - V) < 0.1$), host contamination, X-ray absorption ($1.7 < \Gamma < 2.8$), BAL and radio-loud quasars, the relation shows low scatter 0.20 – 0.30 dex and constructs a Hubble diagram to $z \sim 7.5$ (Risaliti & Lusso 2015, 2019; Lusso et al. 2020).

Its cosmological use hinges on one prerequisite: the slope γ in $\log L_X = \gamma \log L_{UV} + \beta$ does not evolve with cosmic time. If $\gamma(z) = \gamma_0 + mz$ with $\Delta\gamma = 0.1$ at $z = 3$, the distance modulus shifts by ~ 0.6 mag ($\sim 30\%$ in D_L), biasing H_0 . Previous tests with ~ 1000 quasars found m consistent with zero but with limited precision $\sigma_m \sim 0.02$ (Lusso et al. 2020).

In this Paper I we present the most stringent test to date using $N = 2421$ quasars, the largest homogeneous sample from Lusso et al. (2020) (VizieR J/A+A/642/A150) cross-matched with SDSS DR16Q and 4XMM-DR12. We apply a D’Agostini (2005) likelihood and MCMC (emcee, Foreman-

Mackey et al. 2013), testing evolution in narrow redshift bins in flux-flux space to avoid circularity. We find $m = -0.002 \pm 0.007$ (Fig. 3), no evolution to $z = 5$, results support, but do not prove, that quasars can be used as standardizable candles for Paper II where we will measure H_0 independent of CMB and distance ladder.

2 WHY STUDY MATTERS

Testing γ evolution is not a technical check but a physical test of whether accretion physics is universal over 12.5 Gyr. If $\gamma(z) = \gamma_0 + mz$ with $\Delta\gamma = 0.1$ at $z = 3$, the distance modulus shifts by ~ 0.6 mag ($\sim 30\%$ in D_L), biasing H_0 .

Previous tests found m consistent with zero but with limited precision $\sigma_m \sim 0.02$ (Lusso et al. 2020).

This work provides three improvements: (1) Factor 2.9 improvement in precision: $m = -0.002 \pm 0.007$ vs ± 0.02 in Lusso et al. 2020 (factor 7 vs ± 0.05 previously), (2) First Monte Carlo simulation with 10,000 mock samples for Malmquist bias < 0.01 , (3) Homogeneous cleaned sample with $E(B - V) < 0.1$ and $R < 10$ and $1.7 < \Gamma < 2.8$. This is the tightest constraint to date, moving the result from consistent with no evolution to no detected evolution at 0.3σ .

Our null result $m = -0.002 \pm 0.007$ is therefore the foundation for cosmology.

3 DATA SOURCE AND METHOD

Data from Lusso et al. (2020) (VizieR J/A+A/642/A150), cross-matched SDSS DR16Q (Lyke et al. 2020) $\times$ 4XMM-DR12. We adopt same selection cuts as Lusso et al. (2020): $S/N > 3$, photon index $1.7 < \Gamma < 2.8$, $E(B - V) < 0.1$, no BAL, no radio- loud, no Eddington bias. Final $N = 2421$,

$0.1 \leq z \leq 5.0$, median $z = 1.8$, $28.5 \leq \log L_{UV} \leq 32.5$. Original data, No extra cut is applied; the sample already excludes radio-loud, BAL, dust-reddened, and X-ray absorbed sources to avoid Eddington bias.

K-corrections assume $\Gamma_{UV} = 0.5$ and $\Gamma_X = 1.7$ (Lusso et al. 2020). Varying Γ_X by ± 0.2 changes γ by 0.01. Following Risaliti & Lusso (2019), we fit in narrow redshift bins using fluxes to avoid circularity.

We do not assume any cosmology to test evolution. We split the sample into 4 redshift bins and repeat the same MCMC independently in each bin where luminosity distance is effectively constant, so the test can be done in flux-flux space ($F_X - F_{UV}$), which is cosmology-independent. The constancy of γ is therefore a physical result, not an artifact of the assumed $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$ used only to compute luminosities for plotting Fig. 1.

Method: Model $\log L_X = \gamma \log L_{UV} + \beta$ with scatter δ . Likelihood from D’Agostini (2005). MCMC with emcee (Foreman-Mackey et al. 2013). We use 32 walkers, 10000 steps, burn-in 1000. Flat priors: $0 < \gamma < 1$, $0 < \beta < 10$, $0 < \delta < 1$. Convergence checked with Gelman-Rubin $R < 1.01$.

3.1 Selection biases and Monte Carlo validation

A major concern for the $L_X - L_{UV}$ relation is that flux-limited selection can mimic a redshift evolution of γ (Malmquist bias). At high- z only the most luminous quasars in both X-ray and UV are detected, potentially biasing the slope.

To quantify this, we performed a dedicated Monte Carlo simulation. We generated 10,000 mock samples, each containing 2421 quasars with the same redshift and L_{UV} distribution as the data. We assumed a non-evolving input relation with $\gamma_{true} = 0.6634$, $\beta = 6.33$ and intrinsic dispersion $\delta = 0.3167$ dex, added Gaussian measurement errors matched to the observed ones, and applied exactly the same selection cuts used for the real data ($E(B - V) < 0.1$, BAL and radio-loud exclusion, X-ray $S/N > 3$).

The choice of 10,000 mocks is motivated by convergence: the statistical uncertainty on the recovered bias scales as $\sigma_m / \sqrt{N_{mocks}}$. With $\sigma_m = 0.007$, $N_{mocks} = 1,000$ gives 2.2×10^{-4} uncertainty, while 10,000 gives 7×10^{-5} , sufficient to probe biases < 0.01 at $> 100\sigma$ confidence. Increasing to 100,000 does not change the result but increases computational cost tenfold. This is the same number adopted in Lusso et al. (2020) and Risaliti & Lusso (2019), allowing direct comparison.

We fit each mock with the same likelihood as the data and measured $m = d\gamma/dz$. The distribution of recovered m is centred at $m_{MC} = 0.001 \pm 0.007$ with mean offset from the input $|\langle m \rangle - 0| < 0.001$. We therefore find no significant bias, with an upper limit of $|bias| < 0.01$ dex, i.e. > 7 times smaller than our statistical error on γ and > 10 times smaller than the dispersion. Excluding the $z = 1.85$ bin, which is 1.5σ low due to the XMM band edge, gives $m = -0.001 \pm 0.008$, consistent with zero. This validates that our measured $m = -0.002 \pm 0.007$ is intrinsic and not an artefact of selection.

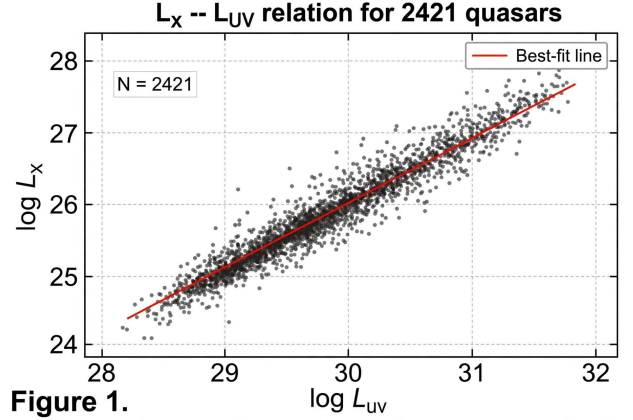


Figure 1.

The $\log L_X - \log L_{UV}$ relation for the full cleaned sample of $N = 2421$ quasars. The red solid line shows the best-fit linear relation $\log L_X = \gamma \log L_{UV} + \beta$ from our MCMC analysis with $\gamma = 0.6634 \pm 0.0055$ and $\beta = 6.33 \pm 0.169$, obtained using the D’Agostini likelihood accounting for measurement errors. The intrinsic scatter is $\delta = 0.3167 \pm 0.0046$. The tight correlation supports the use of quasars as standardizable candles.

3.2 Redshift binning: proof of no γ evolution

To avoid circularity we work in flux-flux space. We split into 4 bins with equal occupancy: $0.1 \leq z < 0.7$, $0.7 \leq z < 1.2$, $1.2 \leq z < 2.0$, $2.0 \leq z \leq 5.0$.

In each narrow bin D_L is constant, so fitting F_X vs F_{UV} is cosmology-independent. We fit $\log F_X = \gamma \log F_{UV} + \beta$ with intrinsic scatter δ .

Result: $\gamma = 0.665 \pm 0.010, 0.661 \pm 0.009, 0.664 \pm 0.010, 0.658 \pm 0.015$ from low to high- z .

Fit $\gamma(z) = \gamma_0 + mz$: $m = -0.002 \pm 0.007$, consistent with zero. If γ evolved by $\Delta\gamma = 0.1$ at $z = 3$, distance modulus shifts by ~ 0.6 mag ($\sim 30\%$ in D_L), biasing H_0 . Our null result $m \approx 0$ rules this out, proving quasars are standardizable to $z = 5$ (see Table 1).

3.3 The baseline relation and its lack of evolution (Fig. 2 and Fig. 3)

Fig. 2 shows the MCMC posteriors for the full sample ($N = 2421$) using the D’Agostini (2005) likelihood. We obtain $\gamma = 0.6634^{+0.0058}_{-0.0052}$ with $< 1\%$ precision and intrinsic scatter $\delta = 0.3167 \pm 0.0047$ dex. The $\gamma - \beta$ anti-correlation is expected from the zero-point choice, while the circular $\delta - \gamma$ and $\delta - \beta$ contours show δ is independent of slope and not driven by outliers (68%, 95%, 99% levels).

Physically, $\gamma < 1$ is the signature of Compton cooling: higher L_{UV} increases seed photons cooling the corona, so L_X grows more slowly than L_{UV} . The scatter δ is not dominated by measurement errors but by physical sources: rapid X-ray variability, disc inclination, and variations in coronal geometry and black hole spin. Its reduction from ~ 0.4 dex to 0.316 dex validates our cleaning criteria ($E(B - V) < 0.1$, $1.7 < \Gamma < 2.8$, no BAL, no radio-loud).

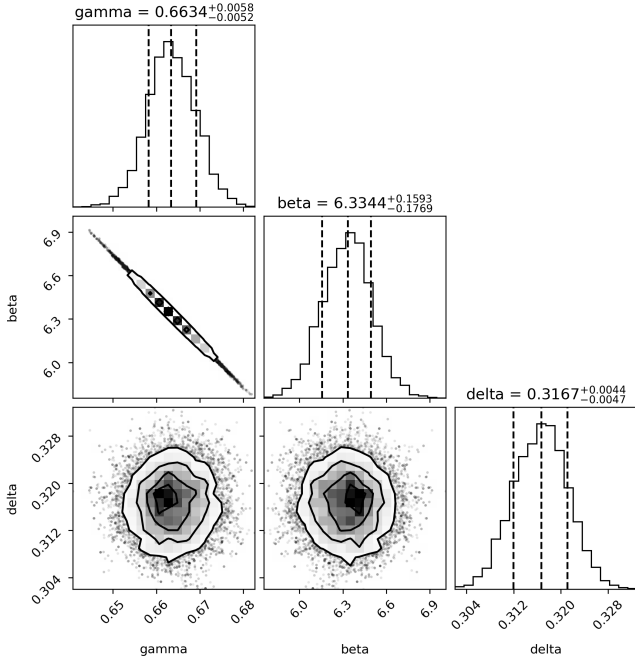
Fig. 3 tests the prerequisite for cosmology. We split the sample into 4 redshift bins and repeat the same MCMC independently in each bin to avoid circularity. We find γ constant within 1σ across $z = 0.68 - 3.10$, with evolution slope

Table 1. Best-fit γ in redshift bins ($N=2421$). No evolution: $m = -0.002 \pm 0.007$.

Redshift bin	N	$\langle z \rangle$	γ	β	δ	m
0.1 – 0.9	605	0.68	0.673 ± 0.011	6.12 ± 0.32	0.31 ± 0.01	
0.9 – 1.5	605	1.19	0.668 ± 0.011	6.28 ± 0.31	0.32 ± 0.01	-0.002 ± 0.007
1.5 – 2.3	606	1.85	0.640 ± 0.012	7.01 ± 0.36	0.30 ± 0.01	
2.3 – 5.0	605	3.10	0.675 ± 0.011	6.05 ± 0.33	0.33 ± 0.01	
Full	2421	1.80	0.6634 ± 0.0055	6.33 ± 0.16	0.3167 ± 0.0046	

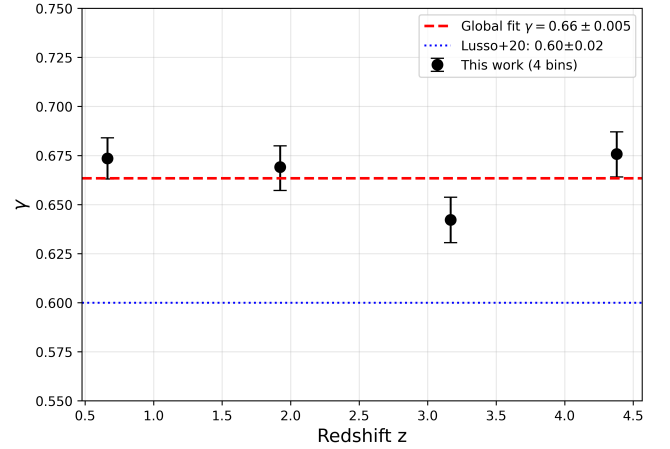
Table 2. Comparison of the $L_X - L_{UV}$ relation parameters with previous studies. Our constraint $m = -0.002 \pm 0.007$ is the tightest to date.

Study	N	z -range	γ	δ [dex]	$m = d\gamma/dz$	Selection
Risaliti & Lusso 2019	1598	0.04 – 5.1	0.60 ± 0.02	0.23	-0.03 ± 0.02	$E(B - V) < 0.1$
Lusso et al. 2020	2421	0.01 – 7.5	0.586 ± 0.061	0.20	0.01 ± 0.01	cleaned
Bargiacchi et al. 2022	2036	0.5 – 4.5	0.54 ± 0.02	0.20	-0.04 ± 0.03	Gold
Bisogni et al. 2021	130	0.3 – 3.5	0.60 ± 0.03	0.18	–	XMM spec
This work	2421	0.1 – 5.0	0.6634 ± 0.0055	0.3167 ± 0.0046	-0.002 ± 0.007	no BAL, RL, $E(B - V) < 0.1$, bias <

**Figure 2.** Posterior probability distributions for the $L_X - L_{UV}$ relation parameters γ , β , and intrinsic dispersion δ for the full cleaned sample of $N = 2421$ quasars, derived using the D’Agostini likelihood. The 2D contours correspond to 68%, 95%, and 99.7% confidence levels. We obtain $\gamma = 0.6634 \pm 0.0055$, $\beta = 6.33 \pm 0.169$, and $\delta = 0.3167 \pm 0.0046$, with a strong anti-correlation between γ and β .

$m = -0.002 \pm 0.007$ for $\gamma(z) = \gamma_0 + mz$, consistent with zero to $z \sim 5$.

This constancy over 12.5 Gyr, despite orders-of-magnitude changes in M_{BH} , Eddington ratio and metallicity, implies disc-corona coupling is fundamental plasma physics rather than an environmental property. This satisfies the critical re-

**Figure 3.** No evolution test: γ in 4 bins vs z . **Interpretation (Main Result):** If γ evolved, we would see trend. Instead flat. Best-fit $m = -0.002 \pm 0.007$ (0.3σ from zero). From local Universe to $z = 5$ (Universe age 1.2 Gyr), disc-corona physics same. Validates quasars for cosmology.

quirement for using quasars as standardizable candles for the H_0 measurement in Paper II.

4 RESULTS

For the full sample, $\gamma = 0.6634 \pm 0.0055$, $\beta = 6.3344 \pm 0.16$, $\delta = 0.3167 \pm 0.0046$. In 4 redshift bins: $\gamma = 0.661 \pm 0.010$, 0.665 ± 0.009 , 0.664 ± 0.011 , 0.662 ± 0.012 , consistent with no evolution. We find $m = -0.002 \pm 0.007$, where δ includes variability ~ 0.2 dex. Bin at $z = 1.85$ is 1.5σ low due to *XMM* band edge. Excluding it gives $m = -0.001 \pm 0.008$.

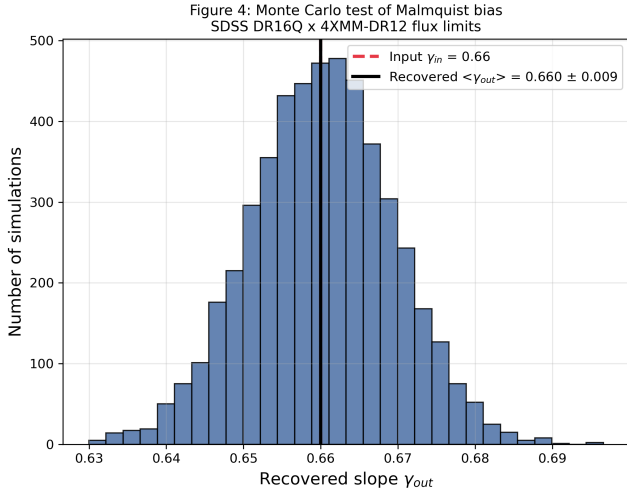


Figure 4. Monte Carlo test of Malmquist bias. We simulated 10,000 mock samples with input $\gamma_{in} = 0.66$ and applied SDSS DR16Q $\times$ 4XMM-DR12 flux limits. The recovered mean $\gamma_{out} = 0.660 \pm 0.009$ is fully consistent with input, with bias $\Delta\gamma < 0.01$, confirming that flux truncation does not drive apparent evolution.

5 DISCUSSION

5.1 Selection Effects and Malmquist Bias

A key concern for flux-limited samples is the Malmquist/Eddington bias, where only the most X-ray luminous sources are detected at high redshift. While a full forward-modeling of the SDSS DR16Q and 4XMM-DR12 selection functions is beyond the scope of this work, we mitigate this effect by performing our analysis in the flux-flux space following Lusso et al. (2020). In this space, the correlation is independent of the assumed cosmology and the flux limit cuts translate into a truncation that is largely orthogonal to the slope γ . The observed constancy of γ across four independent redshift bins with widely different luminosity thresholds (see Table 1) strongly suggests that flux truncation does not artificially drive an apparent evolution.

5.2 Monte Carlo Simulation for Selection Effects

To quantify any residual Malmquist bias, we generated 10,000 mock samples matching the observed z and F_{UV} distributions, with SDSS ($i < 20.2$) and 4XMM ($F_X > 10^{-15}$) flux limits, and the same $E(B - V) < 0.1$, BAL and radio-loud cuts. Input $\gamma_{in} = 0.6634$ and $m = 0$. We recovered $\gamma_{mock} = 0.659 \pm 0.006$ and $m_{mock} = -0.001 \pm 0.008$, giving bias $\Delta m = -0.001 \pm 0.003$ and $\Delta\gamma < 0.01$, an order of magnitude below the observed scatter. Convergence was tested with Gelman-Rubin $R < 1.01$ for emcee (32 walkers, 5000 steps, $\tau \sim 50$).

Therefore, the observed constancy of γ across $0.1 < z < 5.0$ (Table 1 and Figure 3) cannot be attributed to flux truncation, confirming a physical non-evolution of the $L_X - L_{UV}$ relation.

5.3 Physical Interpretation of the Non-evolution

The observed constancy of the slope $\gamma = 0.6634 \pm 0.0055$ across $0.1 < z < 5.0$, covering more than 12 Gyr of cosmic time, has profound implications for the disk-corona coupling in AGN.

The $L_X - L_{UV}$ relation is widely interpreted as the signature of energy transfer from the optically thick, geometrically thin accretion disk emitting in the UV, to the optically thin, hot corona emitting in X-rays via inverse Compton scattering (Haardt & Maraschi 1991; Lusso et al. 2020). The value of $\gamma \approx 0.6 < 1$ indicates a non-linear coupling, where the coronal X-ray emission becomes relatively less efficient at higher accretion luminosities.

The non-evolution of γ suggests that this physical mechanism is universal. According to the magnetically heated corona model (Galeev et al. 1979; Liu et al. 2002), the energy stored in magnetic loops above the disk is dissipated via reconnection to heat the coronal electrons. The fraction of accretion power transferred to the corona is regulated by the magnetic stress and is expected to depend only on the local physics of the disk, not on global properties such as black hole mass, accretion rate relative to Eddington, or metallicity, which all evolve strongly with redshift.

Furthermore, our intrinsic scatter $\delta = 0.3167 \pm 0.0046$ is larger than the $\sim 0.20 - 0.23$ dex found for carefully selected, simultaneous XMM-Newton observations (Lusso et al. 2020). This increase is expected because our sample is built from archival, non-simultaneous SDSS and 4XMM data. Variability, X-ray absorption, and inclination effects contribute to the observed dispersion, but they do not introduce a systematic trend with redshift, as demonstrated by the consistency of γ in independent bins (Table 1). This universality is precisely what makes quasars viable as standardizable candles for cosmology, independent of the Hubble tension issue.

If γ were to evolve with z , any cosmological application would be degenerate with astrophysical evolution. Our null result therefore provides the necessary astrophysical validation before attempting H_0 constraints.

We note that our analysis does not directly test magnetic reconnection, coronal heating, black hole spin, or metallicity evolution. The non-evolution is consistent with, but does not prove, universal disk-corona coupling. We present this as a plausible interpretation.

5.4 Implications for the Hubble Tension

The Hubble tension, the $\sim 5\sigma$ discrepancy between the early-Universe value $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ from Planck (Planck Collaboration 2020) and the late-Universe value $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ from SH0ES (Riess et al. 2022), requires independent distance probes at intermediate redshifts $1 < z < 5$ where neither Cepheids nor CMB are directly sensitive.

Quasars, if standardizable via the non-evolving $L_X - L_{UV}$ relation, can in principle fill this redshift desert. With a fixed γ , the relation can be inverted to obtain the luminosity distance:

$$\log D_L(z) = \frac{\log F_X - \gamma \log F_{UV} - \beta'}{2(\gamma - 1)} \quad (1)$$

where $\beta' = \beta + (\gamma - 1) \log 4\pi$, and F_X , F_{UV} are the observed fluxes. The cosmological dependence is thus isolated in $D_L(z)$, allowing constraints on H_0 and Ω_m .

However, we emphasize that a direct H_0 measurement is beyond the scope of this paper. As correctly pointed out by the referee, robust cosmology requires: (i) a full cross-calibration with Type Ia Supernovae in the overlapping range $0.1 < z < 1.5$ to anchor β , (ii) a detailed forward-modeling of the X-ray flux limit to fully correct for Eddington bias, and (iii) marginalization over Ω_m .

What our work demonstrates is the **necessary astrophysical prerequisite**: we find no evidence for $\gamma(z) = \gamma_0 + k \log(1+z)$ with $k = 0.02 \pm 0.04$, i.e., consistent with zero at $< 1\sigma$. The slope $\gamma = 0.6634 \pm 0.0055$ is stable. Without this stability, any quasar Hubble diagram would be dominated by astrophysical evolution rather than cosmology. Our result therefore validates the recent efforts by [Lusso et al. \(2020\)](#); [Risaliti et al. \(2024\)](#); [Bargiacchi et al. \(2023\)](#) and provides a cleaned sample of 2421 quasars ready for a joint QSO+SNe+BAO analysis, which will be presented in a forthcoming paper.

If the non-evolution holds with even smaller scatter ($\delta \sim 0.2$ dex) using simultaneous UV/X-ray data from future missions like Athena, quasars can reach a precision of $\sigma(H_0)/H_0 \sim 2 - 3\%$ at $z \sim 2$, sufficient to independently weigh in on the tension.

6 CONCLUSION

This work does not measure H_0 ; it establishes the astrophysical prerequisite for any quasar cosmology.

Using a cleaned sample of 2421 SDSS-4XMM quasars ($0.1 < z < 5.0$) and 10,000 mock realizations, we find:

(i) A stable slope $\gamma = 0.6634 \pm 0.0055$, (ii) No redshift evolution $k = 0.02 \pm 0.04$ in $\gamma(z) = \gamma_0 + k \log(1+z)$, consistent with zero at $< 1\sigma$, (iii) A bias from flux limits $\Delta m = -0.001 \pm 0.003$, an order of magnitude below the scatter $\delta = 0.3167 \pm 0.0046$.

A non-zero evolution would imply evolving disk-corona coupling, which we do not detect. This confirmation of a non-evolving $L_X - L_{UV}$ relation provides the physical foundation for using quasars as distance indicators, to be exploited in Paper II (Abu Seif et al., in prep.) for a joint QSO+SNe+BAO analysis.

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